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SUPERSEDING
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MILITARY SPECIFICATION

REFRIGERATORS, MECHANICAL, EXPLOSION-PROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers explosion-proof self-contained refrigerators with electric-motor-driven condensing units.

1.2 Classification. Explosion-proof refrigerators covered by this specification shall be of the following sizes and grades, as specified (see 6.2):

Size 5 - 4.5 cubic feet minimum capacity
Size 6 - 6.0 cubic feet minimum capacity
Size 9 - 9.0 cubic feet minimum capacity
Size 13 - 12.5 cubic feet minimum capacity

Grade A - Outer shell - Baked enamel on carbon steel or aluminum
Inner liner - Aluminum or baked enamel or porcelain enamel on carbon steel

Grade B - Outer shell - Porcelain enamel on carbon steel
Inner liner - Porcelain enamel on carbon steel

Grade C - Outer shell - Corrosion-resisting steel
Inner liner - Corrosion-resisting steel

Grade D - Outer shell - Aluminum
Inner liner - Aluminum

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 4110

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SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
HH-I-524	-	Insulation Board, Thermal (Polystyrene)
QQ-C-320	-	Chromium Plating (Electrodeposited)
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packing

MILITARY

MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing Of

STANDARD

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Bureau of Standards (NBS)

Standard for Devices to Permit the Opening of Household Refrigerator Door from the Inside

(Application for copies should be addressed to the Director, National Bureau of Standards, Washington, DC 20234.)

American National Standards Institute (ANSI)

B38.1 - American Standard Method of Computing Food Storage Volume and Shelf Area of Automatic Household Refrigerators

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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National Fire Protection Association (NFPA)

Standard No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

American Society for Testing Materials (ASTM)

B117 - Method of Salt Spray (Fog) Testing

C282 - Methods of Tests for Acid Resistance of Porcelain Enamel

C347 - Reflectivity and Coefficient of Scatter of White Porcelain Enamels

E97 - 45 Degree and 0 Degree Directional Reflectance of Opaque Specimens by Filter Photometry

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The refrigerators delivered under this specification shall be the manufacturer's current standard commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.2 Safety standard. Refrigerators shall be UL tested and listed as conforming to explosion safety requirements of the National Electrical Code, class I, group C&D for hazardous locations.

3.2.1 Certification. Prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the refrigerators he proposes to supply under this specification meet the requirement of the National Electrical Code, class I, group C&D for hazardous locations. Acceptable

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evidence of meeting the requirements of this standard shall be a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the refrigerators offered have been tested and listed as conforming to explosion safety requirements of the National Electrical Code, class I, group C&D, for hazardous locations.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial refrigerator provided the completed item complies with all provisions of this specification.

3.3.1 Plastics. All plastics used in the refrigerator shall not impart significant odor or taste to food or water (see 4.5.5.1). Plastic shall not craze, crack, or permanently distort, from temperature, contamination, incompatibility with materials of construction, or load under conditions of normal use. Plastics shall not show more than a slight stain when tested as specified in 4.5.5.2. Clean reworked material generated from manufacturing operations may be used by the same manufacturer provided parts produced meet all the applicable requirements of this specification.

3.3.2 Insulation. Insulation shall be nonmoisture absorbing material; shall be homogeneous throughout; and shall have "K" factors specified in 3.3.2.2 and 3.3.2.3, using 75 degrees Fahrenheit (F) mean temperature and expressed in terms of British thermal unit (Btu) per hour per square foot per inch of thickness per degree F temperature difference. Loose fill type insulation will not be acceptable.

3.3.2.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene material conforming to HH-I-524.

3.3.2.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.15, and a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.5.16.

3.3.2.3 Semi-rigid. Semi-rigid insulation shall be "B" fiber or finer industrial type fiberglass with a manufacturer's commercial rated "K" factor of not more than 0.250, with a manufacturer's rated density of at least 1.0 but not more than 3.0 pounds per cubic foot, and shall be held together with a thermosetting binder.

3.3.3 Refrigerants. The refrigerants shall conform to type 12, 22, or 502 of BB-F-1421.

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3.4 Design. The refrigerators shall be self-contained units comprising an insulated cabinet, evaporator, and electric-motor-driven condensing unit. The refrigerator shall be either a one or two door model as specified (see 6.2). The interior of the insulated cabinet shall be divided into a general-storage compartment and a freezer storage compartment. Refrigerators with the door shelves shall not tip forward when the door shelves are loaded, when tested as specified in 4.5.15. The refrigerators shall be completely explosion-proof on both the interior and exterior. The temperature of the outer shell shall not be more than 10 F below an ambient temperature of 90 F when the refrigeration system is in operation and the temperature in the storage compartment is being held at the refrigerators lowest operating range when tested as specified in 4.5.1.

3.4.1 Operating temperatures. The refrigerators shall have the following operating characteristics under no-load conditions (see 4.5.2).

(a) The refrigerators shall be capable of producing maximum average air temperatures in the general-storage compartment of 36 F in an ambient of 70 F, 38 F in an ambient of 90 F, and 41 F in an ambient of 110 F for 60 Hertz operation. On 50 Hertz operation, the refrigerator shall be capable of production maximum average air temperatures in the general-storage compartment of 38 F in an ambient of 70 F, 40 F in an ambient of 90 F, and 45 F in an ambient of 110 F.

(b) The average air temperature in the freezer storage compartment of the refrigerator, shall be lower than minus 10 F.

3.4.2 Icemaking. The refrigeration system shall be capable of freezing water in ice trays filled to capacity in a maximum of six hours in an ambient of 90 F (see 4.5.3).

3.4.3 Defrosting. Defrosting of the refrigerators shall be manually or automatically initiated and shall perform according to the operating requirements that follow when tested as specified in 4.5.4. The defrosting operation shall not raise the average temperature of frozen samples more than 6.5 F, and the average air temperature of the general-storage compartment more than 10 F. Allowable temperature increases shall be based on the highest average temperature observed during the defrosting period and the average temperature observed just prior to initiation of defrost. The refrigerator shall not be restored to a cooling operation until the cooling surface is completely defrosted. Normal operation shall resume at any preselected thermostat temperature setting. During a 4-hour recovery period the temperature of the general-storage compartment shall be automatically restored to within 4 F of the original stablized temperature (see 4.5.4.2). After the 4-hour recovery period there shall be no accumulation of frost remaining on the cooling surfaces. The development of a light film of white frost is permissible during the recovery period. Refrozen droplets of water may be adhered to the evaporator, but they shall show signs of having melted during the defrost cycle.

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3.4.4 Physical characteristics. Unless otherwise specified, the maximum overall dimensions of all sizes of refrigerators and the minimum storage capacities and area of shelves of refrigerators shall be as shown in table I (see 6.2). Storage capacities and shelf areas shall be determined as outlined in ANSI Standard B38.1 except that the volumes occupied by shelves, door shelves, evaporators, and enclosed unusable volumes within these parts, and other unusable spaces, shall be deducted from the refrigerated space in determining the storage capacities. An unusable space is defined as one inaccessible to a 1-1/2-inch cube or any refrigerated volume which cannot be used for storage. Each refrigerator shall have a minimum of two ice-cube trays. The maximum width and height shall be exclusive of removable brackets or hardware, except that the maximum height shall include legs, where provided. The depth or width of the refrigerator shall be such as to allow moving the refrigerator through a clear opening 30 inches wide without removing more than the door, hardware, or other readily removable items.

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TABLE I. Refrigerator dimensions and capacities

Sizes	Maximum overall dimensions			Minimum capacities			
	Width (inches)	Height (inches)	Depth (inches)	Total storage vol (cu ft)	Low-temperature-storage and freezer-storage compartment capacity (cu ft)	Net shelf area	Ice tray capacities (lbs water)
5	36	34-1/2	25	4.5	0.66	11	2.8
6	23-3/4	34-1/4	28-1/4	6.0	0.75	11	2.8
9	24	55-3/4	32-1/2	9.0	1.10	12	2.8
13	31	65	32-1/2	12.5	2.92	16	2.8

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3.5 Construction.

3.5.1 Cabinets. The outer shell and inner liner shall prevent water vapor and water from the outside and water from the inside from entering the insulated space during usage and cleaning when tested as specified in 4.5.14. The bottom sheet shall be securely fastened to the outer shell and provide a rigid structure to maintain the insulated vapor seal (see 4.5.14).

3.5.1.1 Framing. Framing, if used, shall be welded steel or aluminum alloy. No wood shall be used for framing.

3.5.1.2 Outer shells. The outer shells of the grades A and B refrigerators shall be carbon steel sheet of a thickness not less than 24 gage (0.0239 inch nominal). When aluminum is provided in grade A the minimum thickness shall be not less than 18 gage (0.040 inch). The grade A refrigerators shall be finished on the outside surface in baked synthetic enamel. The grade B refrigerators shall be finished on the outside surface in porcelain (vitreous) enamel. Unless otherwise specified (see 6.2), the color of the exteriors of grade A and B refrigerators shall be white. The outer shell of the grade C refrigerators shall be polished stainless steel with a thickness not less than 24 gage (0.025 inch nominal). The outer shell of the grade D refrigerators shall be aluminum alloy sheet with a standard mill finish without stains or discoloration and with a thickness not less than 20 gage (0.032 nominal). If the outer shells of refrigerators are stiffened by channels, braces, or other means, the materials shall be of the following minimum thicknesses: carbon steel sheet, 26 gage (0.018 inch nominal); stainless steel sheet, 26 gage (0.0187 inch nominal); aluminum alloy sheet, 24 gage (0.020 inch nominal) with a standard mill finish without stains or discoloration when the aluminum is not coated with an organic finish.

3.5.1.2.1 Porcelain (vitreous) finish. Porcelain (vitreous) enamel finish on metal shall have all coats fused on at a temperature which shall produce a hard, glossy surface. Vitreous enamel shall be in accordance with commercial practices with respect to discoloration, imperfections, chipped spots, fish scale, burn off, crazing, blisters which will mar the appearance of the item or leave the base metal exposed, beads of enamel, glaze, lumps, and scumming. When vitreous enamel is applied to sheet enameling steel, the visible exposed finish shall consist of at least one base coat and one finish coat, except for areas where the finished coat may be wiped to prevent chippage. Elimination of the base coat when titania-opacified porcelain enamel is used will be acceptable provided steel commercially accepted for this purpose is used. The reflectance of white porcelain enamel shall be not less than 75 percent (see 4.5.6.1).

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3.5.1.2.2 Baked synthetic enamel finishes. Baked synthetic enamel finish shall be applied to zinc-coated steel or to steel, phosphate treated. When exposed to a salt spray for 96 hours, the baked synthetic enamel-coated steel shall show no corrosion (see 4.5.7). Baked synthetic enamel exterior finish shall withstand plasticizer migrating from the door gasket without adverse effect on appearance and without softening.

3.5.1.3 Inner liners of general-storage compartments. The inner liner of the grade A refrigerators shall be aluminum alloy sheet with a standard mill finish without stains or discoloration, or carbon steel sheet. The carbon steel sheet inner liners of the grade A refrigerators shall have the porcelain-enamel finish specified in 3.5.1.2.1 on the main storage side. The inner liner of the grade B refrigerators shall be carbon steel sheet with the porcelain-enamel finish specified in 3.5.1.2.1. The inner liner of the grade C refrigerators shall be polished stainless steel sheet. The inner liner of the grade D refrigerators shall be aluminum alloy sheet with a standard mill finish without stain or discoloration. Thickness of the sheet metal shall be the same as the outer shell specified in 3.5.1.2. The porcelain-enamel finish of the grades A and B shall have a class A or better resistance to acid when tested as specified in 4.5.6.2. The unexposed surfaces of carbon steel sheet inner liners shall be treated to resist corrosion. Unless otherwise specified (see 6.2), the color of the inner liners finished with porcelain enamel shall be white. All corners shall be rounded to allow easy cleaning.

3.5.1.4 Inner liners of low-temperature and freezer-storage compartments. The inner liners of the freezer-storage compartment shall be steel sheet finished with baked synthetic enamel or porcelain enamel, stainless steel sheet, aluminum alloy sheet with a standard mill finish without stains or discoloration, corrosion-resistant treated copper-alloy sheet, or galvanized steel sheet coated with zinc and shall be provided with easily-cleaned corners.

3.5.1.5 Exterior-doors. Exterior doors shall be of the overlapping type with square, convex-curved, or beveled jambs. Each door shall be provided with an effective thermal barrier between exterior and interior surfaces. Each door shall be equipped with a means for holding it in a closed position with sufficient force to affect the gasket seal, and a conveniently located handle or foot-pedal for opening the door shall be provided. If a mechanical latch is used for holding the door closed, it shall be adjustable. Door gaskets should touch the cabinet evenly, over their full length and provide an air-tight, joint (see 4.5.8). Compliance with National Bureau of Standards for Devices to Permit the Opening of Household Refrigerator doors from inside is mandatory. Doors shall swing to the right or left as specified (see 6.2). A right-hand swing shall be described as having hinges located on the right side of the door when facing the refrigerator.

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3.5.1.5.1 Inner liners of the exterior doors. The inner liners of the exterior door shall be carbon steel sheet finished with porcelain enamel specified in 3.5.1.2.1, aluminum sheet, stainless steel sheet, or plastic. Plastic door liners, when assembled to outer doors, shall not show any cracks, or crazing when tested under environmental conditions specified in 4.5.11. When plastic door liners are used and the insulation is foamed-in-place polyurethane employing fluorinated hydrocarbons, the plastic shall be acrylonitrile, butadiene and styrene (ABS). Unless otherwise specified (see 6.2), the color of the inner liners of doors shall be white.

3.5.1.5.2 Breaker strips. Breaker strips made of plastic shall be provided for exterior door openings. Plastic breaker strips shall not show any cracks or crazing when tested under environmental conditions specified in 4.5.11. The bottom breaker strips shall not show any cracks or crazing when given a 2-pound, 1/2-inch-spherical-radius-dart or tup impact test as specified in 4.5.12. The bottom breaker strip shall be designed and installed to prevent liquid spillage or defrost water from entering the insulation space when tested as specified in 4.5.13. ABS plastic for breaker strips shall be used when foamed-in-place polyurethane insulation with fluorinated hydrocarbons is used.

3.5.1.6 Insulation installation. The space between the outer shells and the inner liners shall be filled with insulation that is resistant to fungus, rot, and vermin. The insulation provided shall be resistant to and installed to prevent settling.

3.5.1.7 Shelving. The general-storage compartment shall be equipped with shelves of the wire, bar, or reinforced stamped types. Wire, bar, or stamped shelves shall be made of passivated stainless steel; anodized aluminum alloy; steel plated with copper, nickel, and chromium; steel, electroplated or hot dipped in zinc; or steel, hot dipped in tin. Shelves shall not exhibit more than three visible spots of attack in 12 square inches of shelf area after 96 hours exposure in a salt-spray (see 4.5.7). Sliding shelves shall have stops to prevent unintentional complete withdrawal from the refrigerator and to prevent contact damage to the liner when the shelf is pushed into the cabinet.

3.5.1.8 Ice cube trays and grids. Ice cube trays shall be aluminum alloy, or stainless steel. Ice cube grids shall be of corrosion-resistant metal. Each ice cube tray shall have a mechanical, quick-release mechanism for ejection of ice cubes. Supports or shelves shall be provided for ice cube trays in the freezer compartment.

3.5.1.9 Supports. Supports for interior shelves, trays, baffles, and storage vessels shall be provided for the application. In the event of breakage of a support it shall be replaceable without special tools or removal of storage compartment liner. Supports made of metal shall not exhibit more than three pinpoint breakdowns after 96 hours exposure in a salt-spray (see 4.5.7).

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3.5.1.10 Chromium plating. Chromium plating of brass, aluminum, and zinc alloy shall be as indicated in table II. Unless otherwise specified (see 6.2), the regular grade of plating in table II shall be used. Chromium plating on steel shall conform to class 1, type I or II of QQ-C-320, and shall have the thickness specified in table II. Plating shall be uniform, durable, and shall cover all the surface. Visible chromium plating on the exterior of the refrigerator shall be bright or polished.

TABLE II. Chromium plating

Base metal and finish	Minimum thickness (inch)
Chromium plating brass	
Copper	Optional
Nickel	0.0003
Chromium	0.00001
Chromium plating on aluminum, steel, or zinc alloy die castings	
Copper	0.0003
Nickel	0.0003
Chromium	0.000005

3.5.1.11 Trays, baffles, and special storage vessels. Trays, baffles, and special storage vessels shall be constructed of corrosion-resistant materials.

3.5.2 Electrical characteristics. Unless otherwise specified (see 6.2), the refrigerators shall operate on a 120-volt, 60-Hz, single-phase, electrical power source. A three-wire grounding type cord with a three-prong attachment plug and adapter with ground wire shall be provided and shall extend at least 5 feet, but not more than 9 feet beyond the point at which it is attached to the back of the cabinet. Refrigerators designed for 50/60 Hz, electric current operation shall operate satisfactorily at 50 Hz within plus or minus 10 percent of specified voltage. Any special transformers required to meet specified voltage requirements shall be furnished with each refrigerator.

3.5.3 Refrigeration system. The refrigeration system shall be sealed and designed to operate on one of the refrigerants specified in 3.3.3.

3.5.3.1 Condensing unit. The condensing unit shall be an air-cooled type consisting of a motor, compressor, condenser, controls and other necessary equipment.

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3.5.3.1.1 Air-cooled. Air-cooled condensers shall be constructed of brass; copper; aluminum, or steel, hot-tin dipped or otherwise suitably protected on the outside, after fabrication, to resist corrosion. Fins, where used, shall be soldered, welded, or mechanically bonded to the tubing to insure permanent contact. Where the outer shell of the refrigerator or the metal surface of the duct at the rear of the refrigerator is used as an extended condenser cooling surface, the condenser tubes shall be securely bonded to the extended surface.

3.5.3.2 Evaporators. Evaporators shall be gravity type and shall be stainless steel; brass or copper plated with tin, nickel, or chromium; aluminum; or galvanized steel.

3.5.3.2.1 Defrost water. Defrost water shall be collected in a readily cleanable, readily removable vessel, or shall be carried out of the cabinet to evaporating apparatus. Adequate baffling shall be provided to prevent defrost water from dripping in the general-storage compartment. Tubes, pipes, or baffles, used to collect or carry defrost water shall have corrosion-resistant surfaces and shall be readily cleanable unless so positioned as to be self-cleaning by means of condensate entering the device.

3.5.3.3 Refrigerant and oil charge. The assembled refrigeration system shall be completely dehydrated and charged with the proper amount of refrigerant and oil necessary for operation so that no freeze-up occurs during operation (see 4.5.9). Methanol may be used provided its use is approved by the compressor manufacturer and the warranty of the compressor and any other components of the condensing unit is not voided. If a combination liquid and moisture indicator is used, the moisture content shall not exceed 10 parts per million (ppm) (see 4.5.9).

3.5.3.4 Refrigeration control. The refrigeration operating cycle shall be activated by a temperature sensing type control that starts and stops the compressor motor or evaporator fan or both compressor motor and fan. The differential of the temperature control shall be such that the temperature changes in the general-storage compartment shall not exceed 6 F and in the freezer-storage compartment shall not exceed 4 F during no-load cycling operation when tested in accordance with 4.5.2. The temperature control shall be accessible for adjustment and servicing.

3.5.4 Structural soundness. The assembled refrigerator, when completely packed and ready for shipment, shall not be damaged when tested as specified in 4.5.10. Inspection for damage may include all components and shall include inner liners and outer shells, all plastic parts; all interior shelves; condensing unit; cooling unit; freezer-storage compartment; and the refrigerator door. Damage shall include, for example, crazing, cracking, or permanent distortion of plastic and enameled parts; cracking or permanent distortion of metal parts; dislocation of condensing unit, cooling unit, freezer-storage compartment; inner liner, or refrigerator door.

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3.6 Marking for identification. The manufacturer's standard nameplate data shall be placed on all refrigerators according to the manufacturer's standard practice.

3.7 Workmanship. The finished product shall not be fractured, punctured, malformed, dented, or broken. All sharp corners and edges, burrs, and slivers shall be removed. Welding and brazing (where required) shall be complete, thoroughly fused, and shall not have burn holes, cracks, or fractures. Soldering (where applicable) shall be clean and smooth with no pin holes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Intermediate inspection.

4.2.2.1 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made of the process and all items processed.

(a) Welding of bottom sheet as specified in 3.5.1.1.

(b) Treatment of bottom sheets, legs, and framing of grade A and B refrigerators as specified in 3.5.1.2.

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(c) Application of vitreous enamel as specified in 3.5.1.2.1.

(d) Application of zinc coating as specified in 3.5.1.4.

(e) Installation of insulation as specified in 3.5.1.6.

(f) Application of chromium plating as specified in 3.5.1.10.

4.2.2.2 In-process tests.

4.2.2.2.1 Salt spray test. For the salt spray test in 4.5.7, the lot shall be samples of components (see 3.5.1.2.2, 3.5.1.7, 3.5.1.9) of one type offered for inspection at one time. Two specimens shall be used for each test. Failure of any specimen shall be cause for rejection of the lot.

4.2.2.2.2 Plastics test. The plastic tests shall be performed as specified in 4.5.5.1 and 4.5.5.2. Two specimens shall be used for each test. The plastic specimens shall be representative of the plastic used in the refrigerator and shall be taken from the lot of each type of plastic used on refrigerator components (see 3.3.1), offered for inspection at one time. Failure of any specimen shall be cause for rejection of the lot.

4.2.2.2.3 Foamed-in-place insulation test. Prior to each days production, the test specified in 4.5.16 shall be performed. Any corrections necessary shall be made to equipment before commencing where necessary. Failure to comply with the requirements of 3.3.2.2 shall be cause for failure of this test.

4.2.3 End item inspection. The inspection lot shall consist of all the refrigerators of one type, size, and grade, offered for inspection at one time. The sample unit for this inspection shall be one complete refrigerator.

4.2.3.1 Visual examination. Examination shall be made of the item for defects set forth in table III. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished where required	X	
	Type of finish not as specified, or blistered, peeled, chipped, or area of no film or thin film	X	
	Color not as specified		X
	Poor touch up, orange peel, color separation, smudge, fish scale, scratched, stained, overspray, sag imprint or foreign matter imbedded or otherwise impaired		X
	Any area of rust		X
	Burrs, rough or sharp edges, sharp corners, slivers	X	
Design	Characteristic not in accordance with the specification	X	
Construction and workmanship, general (applicable to all assemblies)	Part missing or not specified type or size	X	
	Any part fractured, punctured, dented, malformed, or broken	X	
	Functioning component that is inoperative or will not function as intended	X	
	Not connected or joined as specified	X	
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended	X	
	Component not readily accessible for servicing where required		X
Welding and brazing (where required)	Not type specified, missing, incomplete, through burn holes, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, undercut, not smooth and uniform, scale or flux deposits not removed		X
Soldering (where applicable)	Missing, not adherent, or incomplete	X	
	Not clean (flux or flux residue not removed), not smooth, or pin holes in solder		X

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured, or loose	X	
Assembly	Any component not properly assembled or secured	X	
	Unit perceptibly out of square or alignment	X	
Marking for identification	Missing, incomplete, not legible		X

4.2.3.2 Dimensional examination. Dimensional examination shall be made of the end item to determine compliance with this specification. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. Each sample refrigerator shall be tested as specified in 4.5.8, 4.5.9, and 4.5.13. Failure of a test shall be recorded as a defect. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units. The tests specified in 4.5.1 through 4.5.4.2; 4.5.6, 4.5.10, 4.5.11, 4.5.12, 4.5.14 and 4.5.15 shall be performed on one refrigerator and failure of any test shall be cause for rejection of the lot.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements (see 3.2.1), shall be examined and validated.

4.4 Examination of preparation for delivery. An examination shall be made to determine that preservation and packaging, packing, and marking are in compliance with MIL-R-12323.

4.5 Tests.

4.5.1 Outer shell temperature test. The refrigerator shall be operated in an ambient temperature of 90 F, plus or minus 2 F until the temperatures in

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the storage compartment stabilizes at the refrigerators lowest operating range, plus or minus 2 F. The temperature of the outer shell shall then be measured at five locations (front, back, top, and sides) to determine compliance with 3.4 and 3.5.3.4. Failure to comply shall be cause for rejection of the item.

4.5.2 No-load test. To determine compliance with the operating temperature requirements of 3.4.1 and with the refrigeration control differential requirements of 3.5.3.4, a no-load test shall be performed in accordance with the procedure described in ANSI Standard B38.1. Three thermocouples shall be placed as follows on a diagonal line: One in the lower left rear of the compartment, the second in the center of the compartment, and the third in the upper right hand corner of the compartment in the door shelf compartment 1 inch from the top of the compartment, and 1 inch from the back or recessed portion of the compartment. When there is no door shelf, the third thermocouple shall be in the upper right hand corner of the compartment, 1 inch in from the face of the door liner. Failure of any thermocouple temperature to comply with the requirements of 3.4.1 shall be cause for rejection of the item.

4.5.3 Ice-making test. An ice-making test following the procedure described in ANSI Standard B38.1 shall be performed on the refrigerator to determine compliance with the requirments of 3.4.2. Failure to comply shall be cause for rejection of the item.

4.5.4 Defrost test. The refrigerator shall be tested in accordance with 4.5.4.1 and 4.5.4.2 to determine compliance with 3.4.3. Inability to conform to the requirements of 3.4.3 shall constitute failure of the test.

4.5.4.1 Defrost test conditions. The freezer compartment shall be loaded with test packages and ice trays as described in ANSI Standard B38.1. The refrigerator shall be operated in an ambient of 90 F and a relative humidity between 80 and 85 percent. The thermostat shall be adjusted so that the average cabinet air temperature shall be 38 F, plus or minus 2 F. The refrigerator shall be operated at the systems voltage.

4.5.4.2 Defrost test procedure.

a. Door opening schedule. During the first hour of the test, the general-storage compartment door shall be opened wide 15 times for 10-second periods at 4 minute intervals. During the following 8-hour period, the same procedure shall be repeated during the fourth and seventh hours. Prior to the first set of door openings, the general-storage compartment shall be loaded with ice trays filled with 16 to 24 ounces of 90 F water. The number of ice trays

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shall be determined on the basis of one tray per 2 cubic feet of general-storage volume. A cellulose sponge, approximately 1 inch by 3 inches by 4 inches in saturated size, shall be placed in each tray so the vertical dimension is 3 inches. The trays shall be replaced with others containing a sponge and 16 to 24 ounces of 90 F water immediately following the last set of door openings and prior to the 16-hour stabilization period.

b. Stabilization. Following the door opening schedule, the refrigerator shall stabilize for the next 16 hours without the door being opened.

c. Defrosting. The defrosting of the freezer compartment shall begin immediately following the stabilization period. The temperature of the general-storage compartment and the freezer compartment for the refrigerator shall be taken continuously, or at intervals of not greater than 15 minutes, starting prior to initiation of the defrost cycle and continuing throughout the recovery period. The refrigerator with automatically-initiated defrost systems may be prepared prior to the test so that they will not defrost during the door opening and stabilization portions of the test. The defrosting period is terminated when the temperature of the general-storage compartment has peaked. The trays and sponges shall then be removed and the refrigerator allowed to operate for a 4-hour recovery period.

4.5.5 Plastic test.

4.5.5.1 Odor and taste test. Plastics shall be given the following odor and taste tests to determine compliance with 3.3.1. Failure of either specimen shall be cause for rejection of the lot.

(a) Place the plastic to be tested with approximately 5-cubic centimeters (cc) of distilled water in the bottom of a 16-ounce, wide-mouth glass jar. A pat of fresh salted butter, 1-1/4 by 1-1/4 by 1/4 inches shall be placed in a shallow dish in the jar but not in contact with the water or the plastic being tested. The jar shall then be closed tightly. Another pat of butter shall be placed in a second open glass jar. The two jars shall be kept at 45 F to 50 F for 24 hours. At the end of this period, there shall be no significant odor or taste difference between the two pats of butter.

4.5.5.2 Stain resistance test. To determine compliance with the stain resistance requirement of 3.3.1, the following test shall be performed: Test specimens shall be placed in contact with tomato juice and shall be partially covered by a watch glass for 15 days at 38 F. After cleaning according to the manufacturer's recommendations, any significant stain shall be cause for rejection of the lot.

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4.5.6 Porcelain (vitreous) enamel tests.

4.5.6.1 Reflectance test. To determine compliance with the requirements of 3.5.1.2.1 the porcelain (vitreous) enamel finish shall be tested for reflectance in accordance with ASTM C347 and ASTM E97. Failure to meet the 75 percent reflectance shall be cause for rejection of the item

4.5.6.2 Acid-resistance test. The test of porcelain enamel for resistance to acid to determine compliance with 3.5.1.3 shall be performed using ASTM Test Method C282-61T. Any test area exceeding the class A requirement shall be cause for rejection of the item.

4.5.7 Salt spray test. Metal components shall be subjected to the 96 hour salt spray test in accordance with ASTM Test Method B117 to determine compliance with 3.5.1.2.2, 3.5.1.7, and 3.5.1.9. Failure to comply shall be cause for rejection of the lot.

4.5.8 Door-gasket seal. To determine compliance with 3.5.1.5, the door shall be closed on a strip of paper the width and thickness of US currency and the paper withdrawn with noticeable resistance at any gasket contact with the door closed normally. The test shall be performed at least once at each side and at the top and bottom of the door. If the configuration of the door does not permit this test, a 100-watt, minimum, lamp shall be placed inside the refrigerator and an observation for light showing between the gasket and the seating face made in a darkened room. Failure to meet either of the above requirements shall be cause for rejection of the item.

4.5.9 Refrigeration system moisture test. The refrigerator shall be operated continuously for 24 hours in an ambient of 110 F. The power shall then be shut off and the refrigerator allowed to stabilize for 24 hours in an ambient of 70 F. The refrigerator shall then be operated continuously for 24 hours in the ambient of 70 F and observations made for freeze-ups to determine compliance with 3.5.3.3. The temperature control shall be on the lowest setting during both 24-hour periods of operation. When the combination liquid and moisture indicator is used, it shall indicate "dry" within the limits specified in 3.5.3.3. Any nonconformance shall be recorded as a defect.

4.5.10 Structural soundness test. The refrigerator, completely packed at level A, B, or C as specified, shall be tested for impact and vibration at the specified A, B, or C test level, respectively, of PPP-P-600 to determine compliance with 3.5.4. Following the test, the refrigerator shall be unpacked, operated, and further examination made for damage.

4.5.11 Environmental cracking resistance test. To determine compliance with the exterior door inner liner and the breaker strip environmental cracking resistance requirement of 3.5.1.5.1 and 3.5.1.5.2, the following test shall be performed.

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After the refrigerator has been pulled down from a room ambient to a cabinet temperature of 38 F, plus or minus 2 F, the exterior door inner liner and the breaker strips shall be completely coated with a thin film of a mixture consisting of 50 parts oleic acid and 50 parts cotton seed oil. The door shelves shall be loaded as specified in 4.5.15. The maximum sized filled container that can be accommodated on each door shelf shall be raised 2-inches and dropped onto the shelf bottom surface at the extreme right and left end of each shelf. Then the door shall be slammed shut 12 times at a rate of 2.1 plus or minus 0.1 feet per second. The refrigerators shall be left under refrigeration for a total of 8 days, repeating the container drop and the 12 door slams each 24 hours. After this test period, the exterior door inner liner and breaker strips shall be examined for cracks or crazing. Any cracks or crazing shall be cause for rejection of the item.

4.5.12 Bottom breaker strip impact test. To determine compliance with the breaker strip impact resistance requirements of 3.5.1.5.2, the following test shall be performed on the same refrigerator used for the test in 4.5.11.

The general-storage compartment temperature of the refrigerator shall be 38 F, plus or minus 2 F, under ANSI Standard B38.1 operating conditions. A 2-pound 1/2-inch-spherical-radius hardened steel dart or tup shall be dropped from a height of 30 inches onto the bottom breaker strip of the general-storage compartment as assembled in the cabinet. The dart or tup shall impact the breaker strip only once with each drop. Three non-coincident points at random shall be tested and the test shall be carried out within 30 seconds of door opening.

The freezer-storage compartment temperature of the refrigerator shall be below 8 F, under ANSI Standard B38.1 operating conditions. A 2-pound, 1/2-inch-spherical-radius hardened steel dart or tup shall be dropped from a minimum height of 6 inches onto the bottom breaker strip of the freezer-storage compartment as assembled in the cabinet. The dart or tup shall impact the breaker strip only once with each drop. Three non-coincident points at random shall be tested and the test shall be carried out within 30 seconds of door opening. After completion of the tests, the breaker strips shall be examined for cracks or crazing, and any cracks or crazing shall be cause for rejection of the item.

4.5.13 Bottom breaker strip seal test. To determine compliance with breaker strip seal requirements of 3.5.1.5.2, the following test shall be performed after the test in 4.5.11.

The refrigerator shall be leveled and a dam 1/2 inch high or higher, of a suitable caulking compound, shall be built across the front of the refrigerator to contain water at the bottom and over the rear portion of the lower breaker strip. The refrigerator liner will then be filled with sufficient water, tinted with a

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suitable water soluble dye, up to the top of the dam and allowed to stand 24 hours. The water will then be removed by sponging or other means, the surfaces wiped dry, and the breaker strip removed. Examination of the interior and insulation shall not show any water penetration as evidenced by the dye coloring. Any water penetration shall be recorded as a defect.

4.5.14 Cabinet water-tightness test. The refrigerator shall be operated continuously for a period of 21 days in an ambient temperature of 90 F and a humidity of 85 percent. The internal temperature shall be the lowest setting obtainable. The refrigerator doors shall be closed throughout the test period. At the conclusion of the 21 days of operation, the refrigerator shall be disconnected and allowed to defrost for a period of 6 hours in an ambient temperature of 90 F with the doors in a fully open position. After defrosting, all breaker strips shall be removed and an observation made if moisture has accumulated in the insulation or the insulated space to determine compliance with the cabinet water-tightness requirement of 3.5.1. Any moisture penetration shall be cause for rejection of the item.

4.5.15 Stability. The refrigerator door shelves, when provided, shall be loaded with 12 pound per square foot, equally distributed. The remainder of the refrigerator with all shelves and accessories in place shall not be loaded. The loaded door shall be opened 90 degrees and the refrigerator observed for tipping to determine compliance with 3.4. Any tipping shall be cause for rejection of the item.

4.5.16 Density test for foamed-in-place insulation. One free foamed sample of insulation specified in 3.3.2.2 shall be taken. The sample shall be a cube, 8 inches on a side. A cube, 6 inches on a side shall be cut from the center of the sample and weighed. This weight of the 6-inch cube shall be multiplied by 8 to determine compliance with the density requirement of 3.3.2.2. In addition, a daily record of the foam insulation shall be kept as follows and made available to the contracting officer: Calibration of mix (mix proportion), amount of flow (per box), time of flow (per box). A density of less than 1.7 pound per cubic foot shall be cause for failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Each refrigerator shall be preserved and packaged in accordance with the applicable requirements of MIL-R-12323.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Levels A, B and C. Each refrigerator, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-12323.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-R-12323.

6. NOTES

6.1 Intended use. The refrigerators covered by this specification are for general laboratory use in which materials to be refrigerated may be explosive or in any room with an explosive atmosphere.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and grade required (see 1.2).
(c) Number of exterior doors (one or two) (see 3.4).
(d) Whether dimensions shall be other than specified (see 3.4.4).
(e) Whether colors of exteriors and interiors shall be other than specified (see 3.5.1.2, 3.5.1.3, and 3.5.1.5.1).
(f) Whether doors shall be left or right hand (see 3.5.1.5).
(g) Whether other than regular grade of chromium plating is required (see 3.5.1.10).
(h) Electric system operation if other than specified (see 3.5.2).
(i) Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).

6.3 End item tests. Performance of tests as specified in 4.5.1 through 4.5.4.2, 4.5.6, 4.5.10, 4.5.11, 4.5.12, 4.5.14, and 4.5.15 should be limited to one successful demonstration of compliance to the requirements of this specification for each contract.

6.4 Supersession data. The following changes were made to classification paragraph 1.2:

MIL-R-43592	MIL-R-43592A
Type I - Low-temperature-storage compartment below 32 degrees Fahrenheit (F); manually initiated defrost with manual or automatic-reset.	None

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MIL-R-43592	MIL-R-43592A
Size 5 - 4.5 cubic feet minimum capacity Size 6 - 6.0 cubic feet minimum capacity Size 9 - 9.0 cubic feet minimum capacity	Size 5 - 4.5 cubic feet minimum capacity Size 6 - 6.0 cubic feet minimum capacity Size 9 - 9.0 cubic feet minimum capacity Size 13 - 12.5 cubic feet minimum capacity
Type II - Isolated freezer-storage compartment below 8 F; automatic defrost with automatic-reset for general-storage compartment only. Size 13 - 12.5 cubic feet minimum capacity	None

Custodians:

Army - GL
Navy - YD
Air Force - 82

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 4110-0282

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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